Exercise on Strings

- How many permutations of the letters *ABCDEFGH* contain the string *ABC*?

- **Solution:**
  Since the string ABC must occur together, we can consider it as a single letter (i.e. denote it by S). So now we only need to find the number of permutations of six SDEFGH. The answer is $P(6,6) = 6! = 720$.